

Digit	Contents	Encoding	Notes
1 (left-most)	- Version 0 - VID_PID present flag - 2 ms-bits of discriminator	DIGIT[1] := (VID_PID_PRESENT << 2) (DISCRIMINATOR >> 10)	Allows first digit typed/spoken to determine version and VID/PID present. Yields a decimal number from 0..7 (0..3 if VID,PID not present). First digit of '8' or '9' would be invalid for v1 and would indicate new format (e.g. version 2)
2..6	- 3rd and 4th ms-bits of Discriminator - 14 ls-bits of PASSCODE	DIGIT[2..6] := ((DISCRIMINATOR & 0x300) << 6) (PASSCODE & 0x3FFF)	Yields a 5-digit decimal number from 00000 to 65535 (0xFFFF/16 bits)
7..10	- 13 ms-bits of PASSCODE	DIGIT[7..10] := (PASSCODE >> 14)	Yields a 4-digit decimal number from 0000 to 8191 (0x1FFF/13 bits)
11	- Check Digit	DIGIT[11] := (CHECK_DIGIT)	See Section 5.1.4.1.5, “Check Digit” for encoding

Table 62. Encoding Method with Vendor and Product ID's included (VID_PID_Present == 1)

Digit	Contents	Encoding	Notes
1 (left-most)	- Version 0 - VID_PID present flag - 2 ms-bits of Discriminator	DIGIT[1] := (VID_PID_PRESENT << 2) (DISCRIMINATOR >> 10)	Allows first digit typed/spoken to determine version and VID/PID present. Yields a decimal number from 0..7 (4..7 if VID,PID present). First digit of '8' or '9' would be invalid for v1 and would indicate new format (e.g. version 2)
2..6	- 3rd and 4th ms-bits of Discriminator - 14 ls-bits of PASSCODE	DIGIT[2..6] := ((DISCRIMINATOR & 0x300) << 6) (PASSCODE & 0x3FFF)	Yields a 5-digit decimal number from 00000 to 65535 (0xFFFF/16 bits)
7..10	- 13 ms-bits of PASSCODE	DIGIT[7..10] := (PASSCODE >> 14)	Yields a 4-digit decimal number from 0000 to 8191 (0x1FFF/13 bits)
11..15	- Vendor ID	DIGIT[11..15] := (VENDOR_ID)	Yields a 5-digit decimal number from 00000 to 65535 (0xFFFF/16 bits)